

Two-flavour quartic: the 2|2 partition, one-loop RG-invariant loci, and intersection with the 1|3 entropy-maximal sector

1 Setup

We consider a real two-flavour scalar with quartic coupling tensor $\lambda_{ijkl} = \lambda_{(ijkl)}$ and interaction

$$V = \frac{1}{4!} \lambda_{ijkl} \phi_i \phi_j \phi_k \phi_l, \quad i, j, k, l \in \{1, 2\}. \quad (1)$$

A convenient parametrization of the five independent components is

$$\lambda_{1111} = a, \quad \lambda_{1112} = b, \quad \lambda_{1122} = c, \quad \lambda_{1222} = d, \quad \lambda_{2222} = e. \quad (2)$$

At one loop in $d = 4 - \epsilon$, the beta function is

$$\beta_{ijkl} := \mu \frac{d\lambda_{ijkl}}{d\mu} = -\epsilon \lambda_{ijkl} + \kappa \left(\lambda_{ijmn} \lambda_{mnkl} + \lambda_{ikmn} \lambda_{mnjl} + \lambda_{ilmn} \lambda_{mnjk} \right), \quad (3)$$

with a convention-dependent positive constant κ (for instance $\kappa = 1/(16\pi^2)$ in a standard normalization). In components,

$$\beta_a = -\epsilon a + \kappa(3a^2 + 6b^2 + 3c^2), \quad (4)$$

$$\beta_b = -\epsilon b + \kappa(3ab + 6bc + 3cd), \quad (5)$$

$$\beta_c = -\epsilon c + \kappa(ac + 2b^2 + 2bd + 4c^2 + ce + 2d^2), \quad (6)$$

$$\beta_d = -\epsilon d + \kappa(3bc + 6cd + 3de), \quad (7)$$

$$\beta_e = -\epsilon e + \kappa(3c^2 + 6d^2 + 3e^2). \quad (8)$$

2 The 2|2 flattening

For the bipartition $(ij)|(kl)$, the flattening matrix is

$$M_{2|2}[(ij), (kl)] = \lambda_{ijkl}. \quad (9)$$

In the ordered basis (11, 12, 21, 22) one has

$$M_{2|2} = \begin{pmatrix} a & b & b & c \\ b & c & c & d \\ b & c & c & d \\ c & d & d & e \end{pmatrix}. \quad (10)$$

The antisymmetric two-particle state $|A\rangle = (|12\rangle - |21\rangle)/\sqrt{2}$ is always a null vector, so it is convenient to pass to the symmetric basis

$$|11\rangle, \quad |S\rangle = \frac{|12\rangle + |21\rangle}{\sqrt{2}}, \quad |22\rangle. \quad (11)$$

On this three-dimensional support the flattening becomes

$$M_{\text{sym}} = \begin{pmatrix} a & \sqrt{2}b & c \\ \sqrt{2}b & 2c & \sqrt{2}d \\ c & \sqrt{2}d & e \end{pmatrix}. \quad (12)$$

The $2|2$ reduced density matrix is then

$$\rho_{2|2} = M_{2|2} M_{2|2}^T, \quad (13)$$

or equivalently $\rho_{\text{sym}} = M_{\text{sym}}^2$ on the symmetric support.

A basis adapted to the exchange $|11\rangle \leftrightarrow |22\rangle$ is

$$|+\rangle = \frac{|11\rangle + |22\rangle}{\sqrt{2}}, \quad |0\rangle = |S\rangle, \quad |-\rangle = \frac{|11\rangle - |22\rangle}{\sqrt{2}}. \quad (14)$$

In this basis,

$$\widetilde{M} := \begin{pmatrix} \frac{a}{2} + c + \frac{e}{2} & b + d & \frac{a-e}{2} \\ b + d & 2c & b - d \\ \frac{a-e}{2} & b - d & \frac{a}{2} - c + \frac{e}{2} \end{pmatrix}. \quad (15)$$

This form makes two natural $2|2$ -adapted loci immediate.

3 Natural $2|2$ -adapted invariant loci

3.1 The even block locus $\mathcal{E}_+$

If

$$\mathcal{E}_+ : \quad a = e, \quad b = d, \quad (16)$$

then (15) becomes

$$\widetilde{M}|_{\mathcal{E}_+} = \begin{pmatrix} a + c & 2b & 0 \\ 2b & 2c & 0 \\ 0 & 0 & a - c \end{pmatrix}. \quad (17)$$

Thus the $|-\rangle$ line decouples, while the $(|+\rangle, |0\rangle)$ plane forms a closed block. This is the most obvious $2|2$ -adapted codimension-2 locus.

It is one-loop RG invariant. Indeed,

$$(\beta_a - \beta_e)|_{a=e, b=d} = \kappa 3(a^2 + 2b^2 - 2d^2 - e^2)|_{a=e, b=d} = 0, \quad (18)$$

$$(\beta_b - \beta_d)|_{a=e, b=d} = \kappa 3(ab + bc - cd - de)|_{a=e, b=d} = 0, \quad (19)$$

so the conditions (16) are preserved. The reduced flow on $\mathcal{E}_+$ is

$$\dot{a} = -\epsilon a + \kappa(3a^2 + 6b^2 + 3c^2), \quad (20)$$

$$\dot{b} = -\epsilon b + \kappa 3b(a + 3c), \quad (21)$$

$$\dot{c} = -\epsilon c + \kappa(2ac + 6b^2 + 4c^2), \quad (22)$$

with $e = a$ and $d = b$ throughout.

3.2 The odd block locus $\mathcal{E}_-$

If instead

$$\mathcal{E}_- : \quad a = e, \quad d = -b, \quad (23)$$

then (15) becomes

$$\widetilde{M}|_{\mathcal{E}_-} = \begin{pmatrix} a+c & 0 & 0 \\ 0 & 2c & 2b \\ 0 & 2b & a-c \end{pmatrix}. \quad (24)$$

Now the $|+\rangle$ line decouples, while the $(|0\rangle, |-\rangle)$ plane forms a closed block. This is a second natural $2|2$ -adapted codimension-2 locus.

It is also one-loop RG invariant. Indeed,

$$(\beta_a - \beta_e)|_{a=e, d=-b} = \kappa 3(a^2 + 2b^2 - 2d^2 - e^2)|_{a=e, d=-b} = 0, \quad (25)$$

$$(\beta_b + \beta_d)|_{a=e, d=-b} = \kappa 3(ab + 3bc + 3cd + de)|_{a=e, d=-b} = 0, \quad (26)$$

so the conditions (23) are preserved. The reduced flow on $\mathcal{E}_-$ is

$$\dot{a} = -\epsilon a + \kappa(3a^2 + 6b^2 + 3c^2), \quad (27)$$

$$\dot{b} = -\epsilon b + \kappa 3b(a + c), \quad (28)$$

$$\dot{c} = -\epsilon c + \kappa(2ac + 2b^2 + 4c^2), \quad (29)$$

with $e = a$ and $d = -b$ throughout.

Remark 1. *The full five-dimensional coupling space is of course RG invariant trivially. The significance of $\mathcal{E}_\pm$ is that they are the two simplest nontrivial linear slices singled out by the $2|2$ flattening: in the parity basis they are exactly the loci where one of the parity lines decouples from the remaining two-dimensional block.*

4 Recall: the $1|3$ entropy-maximal manifold

For the $1|3$ split, the reduced density matrix on the one-index side is

$$\rho_{1|3} = \begin{pmatrix} a^2 + 3b^2 + 3c^2 + d^2 & ab + 3bc + 3cd + de \\ ab + 3bc + 3cd + de & b^2 + 3c^2 + 3d^2 + e^2 \end{pmatrix}. \quad (30)$$

Maximal $1|3$ entropy is equivalent to

$$X := a^2 + 2b^2 - 2d^2 - e^2 = 0, \quad (31)$$

$$Y := ab + 3bc + 3cd + de = 0. \quad (32)$$

We denote this entropy-maximal manifold by $\mathcal{M}_{1|3}$.

The $SO(2)$ -symmetric line is

$$\lambda_{\text{singlet}} = s \left(1, 0, \frac{1}{3}, 0, 1 \right), \quad (33)$$

which certainly lies in $\mathcal{M}_{1|3}$.

5 Intersection with the 1|3 entropy-maximal sector

We now intersect the natural 2|2-adapted invariant loci with $\mathcal{M}_{1|3}$.

5.1 $\mathcal{E}_+ \cap \mathcal{M}_{1|3}$

On $\mathcal{E}_+$, i.e. $a = e$ and $d = b$, the first condition (31) is automatic,

$$X|_{\mathcal{E}_+} = a^2 + 2b^2 - 2b^2 - a^2 = 0, \quad (34)$$

while the second becomes

$$Y|_{\mathcal{E}_+} = ab + 3bc + 3cb + ba = 2b(a + 3c). \quad (35)$$

Hence

$$\mathcal{E}_+ \cap \mathcal{M}_{1|3} = \{a = e, d = b, b(a + 3c) = 0\}. \quad (36)$$

This splits into two branches:

$$\mathcal{S}_0 : \quad a = e, \quad b = d = 0, \quad (37)$$

$$\mathcal{B}_+ : \quad a = e, \quad d = b, \quad a = -3c. \quad (38)$$

The branch $\mathcal{S}_0$ is RG invariant, since setting $b = d = 0$ is preserved by (21). By contrast, the branch $\mathcal{B}_+$ is *not* preserved generically. Indeed,

$$\left. \frac{d}{dt}(a + 3c) \right|_{\mathcal{B}_+} = \beta_a + 3\beta_c = \kappa \left(\frac{8}{3}a^2 + 24b^2 \right), \quad (39)$$

which does not vanish except at the origin.

5.2 $\mathcal{E}_- \cap \mathcal{M}_{1|3}$

On $\mathcal{E}_-$, i.e. $a = e$ and $d = -b$, both 1|3 conditions are automatic:

$$X|_{\mathcal{E}_-} = a^2 + 2b^2 - 2(-b)^2 - a^2 = 0, \quad (40)$$

$$Y|_{\mathcal{E}_-} = ab + 3bc - 3cb - ba = 0. \quad (41)$$

Therefore

$$\mathcal{E}_- \subset \mathcal{M}_{1|3}. \quad (42)$$

Since $\mathcal{E}_-$ is already RG invariant, it follows that $\mathcal{E}_-$ is a two-parameter RG-invariant subspace entirely contained inside the 1|3 entropy-maximal manifold.

6 Does the intersection single out the $SO(2)$ line?

No. The $SO(2)$ -symmetric line (33) belongs to both $\mathcal{E}_+$ and $\mathcal{E}_-$, because there $b = d = 0$ and $a = e$. However, the intersection of the natural 2|2-adapted invariant sector with the 1|3 entropy-maximal RG-invariant sector is much larger:

$$\mathcal{E}_- = \{(a, b, c, d, e) = (a, b, c, -b, a)\} \quad (43)$$

is itself entirely contained in $\mathcal{M}_{1|3}$ and preserved by the one-loop RG flow. Hence requiring

- maximal 1|3 entropy,
- one-loop RG invariance,
- and membership in a natural 2|2-adapted invariant locus,

does *not* single out the $SO(2)$ line. One still has the full two-parameter family $\mathcal{E}_-$.

Equivalently, the strengthened criterion based on combining the two inequivalent bipartitions is still not strong enough to force symmetry. The $SO(2)$ locus remains a distinguished one-dimensional sublocus inside a larger RG-invariant, partition-compatible family, but it is not isolated by these requirements alone.

7 Summary

For the two-flavour symmetric quartic, the 2|2 flattening naturally suggests two codimension-2 linear loci:

$$\mathcal{E}_+ : a = e, b = d, \quad \mathcal{E}_- : a = e, d = -b, \quad (44)$$

corresponding respectively to decoupling of the $|-\rangle$ and $|+\rangle$ parity lines in the 2|2 symmetric basis. Both are preserved by the one-loop RG flow. Their intersections with the 1|3 entropy-maximal manifold behave differently:

- $\mathcal{E}_+ \cap \mathcal{M}_{1|3}$ contains an invariant branch $\mathcal{S}_0$ and a non-invariant branch $\mathcal{B}_+$;
- $\mathcal{E}_-$ lies entirely inside $\mathcal{M}_{1|3}$ and is fully RG invariant.

Therefore the intersection of 2|2-adapted one-loop invariant structure with the 1|3 entropy-maximal RG-invariant sector still does not isolate the $SO(2)$ line.